

How to think in writing

The thought behind the thought



HENRIK KARLSSON

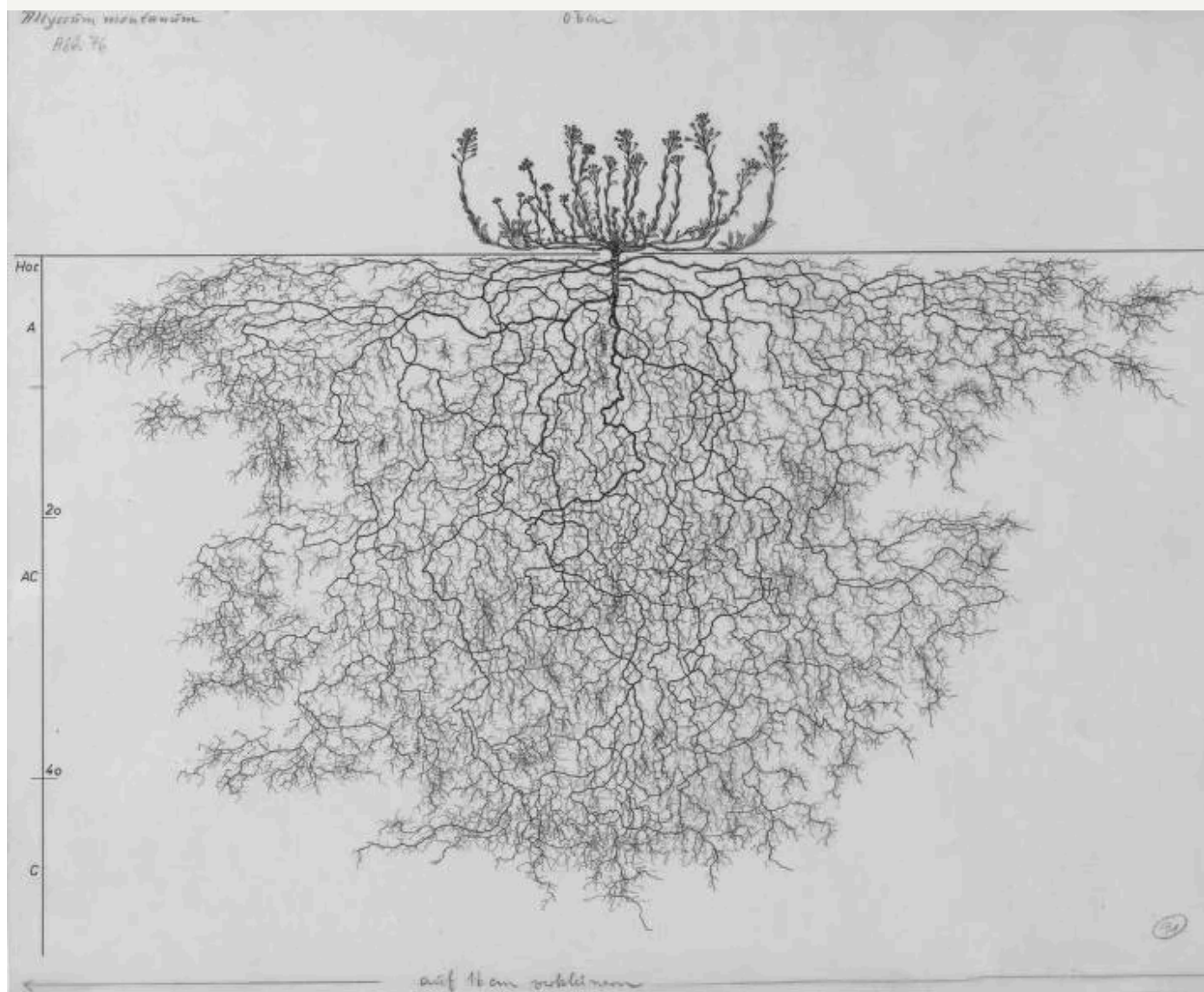
APR 25, 2024

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The reason I've spent so long establishing this rather obvious point [that writing helps you refine your thinking] is that it leads to another that many people will find shocking. If writing down your ideas always makes them more precise and more complete, the

no one who hasn't written about a topic has fully formed ideas about it. And someone who never writes has no fully formed ideas about anything nontrivial.

It feels to them as if they are using their own thinking to write words that you don't understand. You'll not only not

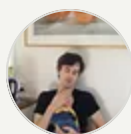
—Paul Graham

When I sit down to write, I feel like I'm in a world where satellites pass overhead like dreams; they're not real. When I write, it is as if I pick them up

But it is hard to do. My ideas are weak, or even wrong. I don't know the right way.

Until last fall, I had not seen anyone properly articulate the mental moves that make writing a powerful tool for thought. Writing advice is usually focused on more superficial parts of the craft. Whatever I knew about thinking on the page I had picked up through trial and error and conversations with other writers.

But then I read Imre Lakatos's [Proofs and Refutations](#). It is not, at first glance, a book about writing. It is a book of mathematical philosophy. By a Hungarian Stalinist, no less. But it is, if you read it sideways, a profound exploration of the craft of writing. This shouldn't be a surprise. Mathematics is, after all, a subset of writing—it is a way of crafting a language that helps you express and improve your thoughts. The main difference, compared to prose writers and poets, is that mathematicians are more rigorous. Because of the precision, reading Lakatos gives



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I write essays about relationships, thinking, and agency, sometimes with my wife. We live on a windswept island in the Baltic Sea with our two daughters.

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me a clearer understanding of what I do, or strive to do, as I sit down each morning and grapple with my thoughts.

What follows is a series of meditations about thinking in writing provoked by, I am not faithful to, Lakatos's book.

A warning. If you aim to write and publish stuff, this essay might tie you up in knots. It is about thinking, not about crafting beauty or finishing things in a first time.

Setting yourself up for defeat

There is a crack, a crack in everything

That's how the light gets in.

—Leonard Cohen, “Anthem”

In [a recent interview with Dwarkesh](#), Patrick Collison explained the value of writing using a metaphor I enjoyed:

Bruno Latour spoke about how he thinks the printing revolution, like Gutenberg's, partially caused the scientific revolution by *making knowledge more rigid*. Before, if some observation didn't match some claim, you could always shrug and be like: “Well, the person who transcribed that thing made a mistake.” So *by making things more rigid, it's easier to break them*. [Emphasis mine]

Good thinking is about pushing past your current understanding and reaching thought behind the thought. It often requires breaking old ideas. This is easier to do when the ideas are as rigid as they get on the page. In a fluid medium like thought or conversation, you can always go, “Well, I didn't mean it like *that*” or

rely on the fact that your short-term memory is too limited for you to notice the contradiction between what you are saying now and what you said 12 minutes ago.

When I write, I get to observe the transition from this fluid mode of thinking to the rigid. As I type, I'm often in a fluid mode—writing at the speed of thought. I feel confident about what I'm saying. But as soon as I stop, the thoughts solidify, rigid on the page, and, as I read what I've written, I see cracks spreading through my ideas. What seemed right in my head fell to pieces on the page.

Seeing your ideas crumble can be a frustrating experience, but *it is the point* if you are writing to think. You want it to break. It is in the cracks the light shines in.

When I write, I push myself to make definite positive claims. Ambiguity allows thought to remain fluid on the page, floating into a different meaning when pushed under pressure. This makes it harder to push your thinking deeper. By making clear and sharp claims, I reveal my understanding so that I—or the person I'm writing to—can see the state of my knowledge and direct their feedback to the point where it will help my thinking improve.

This is valuable to do even in areas where you know way too little to “warrant” an opinion. I met a Japanese linguist in the harbor yesterday and talked about the relationship between the Chinese and the Japanese writing systems. This is a topic I had thought about for about twenty seconds before this. “So,” I said after two minutes, “this is a stupid question, but is the relationship between China and Japan like that between Ancient Greece and the Roman Empire?” This is, as it turns out, not a good analogy. But by spelling out my naive understanding, I gave the linguist a good area to work on when he laid out a richer model of the flow of cultural influence in East Asia.

In the terminology of mathematics, what I did here (and in my writing) was to “make a conjecture,” a qualified guess based on limited information. A hypothesis

The mathematician Alexander Grothendieck, whom Johanna and I [have written about elsewhere](#), would always summarize his first impression of a new situation with a conjecture, proclaiming with irrepressible enthusiasm, “It must be true!” Ten seconds later, someone would come up with a counterexample that proved him wrong. But being right wasn’t the point: getting a better understanding was. And he would immediately throw out a new conjecture. (Holden Karnofsky has a blog post about using this technique to [learn through writing](#).)

Forcing the diffuse ideas and impressions in your head into a definite statement is an art form. You have to grab hold of what is floating and make it rigid and sharp. It can feel almost embarrassing, revealing your ignorance with as much vulnerability as possible.

And it is only the first step. Once you have made your thoughts definite, clear, concrete, sharp, and rigid, you also want to stretch them thin.

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Spreading the frontline until it snaps

By stretching thin I mean “interrogating the conclusion to come up with an explanation of why it *could* be true.” What premises and reasoning chains lead to this conclusion? The explanation isn’t meant to prove that your conclusion was right. It is just a way of unpacking it.

By stretching a claim into an explanation, you spread it on a “wider front” (to borrow a metaphor from Lakatos), so that the criticism has more targets.

I used this tactic in the food store yesterday. Maud, our six-year-old, told me we had to get a pink miniature plastic teapot. I couldn't come up with a compassionate counterargument, so I said, “Why do you think a plastic teapot is so great?” And she said, “Because it is so beautiful. And I need one in plastic so it doesn't break. I would use it all the time.” This brought a smile to my face. Seeing her trying to prove her point, she had given me three times as many claims to attack.

Since the goal is to find flaws in our guesses (so that we can change our minds, refine our mental models and our language, and be more right) stretching a claim thin through an explanation is progress. Even if the explanation is wrong.

You are interested only in proofs which ‘prove’ what they have set out to prove.
I am interested in proofs even if they do not accomplish their intended task.
Columbus did not reach India but he discovered something interesting.
—Lakatos

Let me take another example. Before Maud was born, Johanna and I worked as teachers in Sweden. The first conclusion we drew from that experience was that we didn't want to submit our kids to what we had observed. This way of formulating it (“Not *that*”) is a bit vague as it only defines where not to look for a solution. It is useful to also attempt a positive formulation. If I were to reconstruct the positive version of our conclusion back then, it was something like, “We need to find (or start) a school where our daughter can pursue her interests at her pace.”

There are several subtle problems with this conclusion. But the point is—these problems didn't come into view until we had stretched and probed our original position.

The way we stretched and improved our conclusion back then was more haphazard than it would have been today. We just talked about it aimlessly, read randomly, and made small notes. This cost us time and caused confusion. These days, I would instead stretch a conclusion like this as a series of bullet points where I spell out the intuition behind my claim in a series of premises. In the case of Maud's education, this would have looked something like this (note that this is not my current understanding but a reconstruction of what I thought eight years ago):

- People have an intrinsic motivation to learn and it is important to not undermine that, which schools do
- It is better to go deep on a few topics that you are passionate about rather than have a superficial understanding of a broad range of subjects you care little about
- But you need to attend a school so you get socialized
- Hence, we need to find a school that allows self-directed learning

Once I stretch my understanding thin in writing, I often see holes right away. I start correcting myself and discarding ideas already while typing. I cut ideas that are obviously flawed. I rewrite what feels ambiguous to make it more precise, concrete, unhedged, and true to my understanding.

The flaws I see immediately, however, are only the more superficial flaws. The deeper patterns take a longer time to emerge—because they are further from my established thoughts and so are harder to articulate.

Often, they occur first as subtle emotional cues. As I reread a passage, I notice a slight tension across my chest or my eyes fog over. For some reason, it doesn't feel right. There is something wrong here.

These subtle feelings are easy to dismiss (“Eh, words are slippery, I mean something slightly different . . . there is no reason to obsess about this”). But in experience, it is these subtler problems that tend to open a path beyond my current understanding. I learned this from my wife, Johanna, who will often sit with a draft for several hours, not writing or editing, but simply articulating what something feels off to her. Our best essays have come out of the things she surfaced during those sessions.

For this reason, I suspect that many of my friends who write and publish rapidly are shortchanging themselves. They generate texts filled with hidden doors and move on before they’ve opened them.

I tend to go through my list of premises and assumptions and ask follow-up questions to myself, to further extend my conclusion. To continue the example from above, I would take one of the premises and stretch it out like this:

- But you need a school so you get socialized
 - *Curious: why?*
 - Kids will get depressed and struggle to navigate workplaces, and so on, they haven’t been exposed to society
 - *Where can I read more about this? Are there any good studies?*
 - Being in something like a school is important because humans are social animals. We pick up most of our skills and norms and so on by being immersed in a peer group
 - *And what follows from this?*
 - *If we are shaped by our peer group, what would the ideal peer group look like?*

The emotional tone of these questions is, in my head, lovingly curious; I’m not trying to put myself down. I’m out to kill ideas. I want to help them evolve and

spill forth more insight. Often this dialogue ends with me changing my mind about several premises and coming to a different conclusion, but the original idea remains the seed—no less valuable for having been proven wrong. It takes creativity and boldness to leap out and form a conclusion, and the part that criticizes must understand how dependent it is on the part that throws ideas at wall. It is often easier to criticize than it is to synthesize a new position.

Counterexamples, local and global

The sun is above the horizon now, the satellites hid behind a thin layer of orange and pink. A hare raises on his hind legs in the middle of the meadow looking around. I tap the glass and watch his ears turn my way.

Now that I have spelled out my position and fixed the obvious flaws, I start probing myself more seriously to see if I can get the argument to break down.

If one of the premises I have unpacked is a factual claim, I'll spend a few minutes skimming research in the area to see how well my position holds up. "Oh, it turns out that most homeschooled kids do *not* have any problems with socialization!" I realized when doing this in relation to Maud's education. (Though it didn't take me a few minutes, it took me years in this case. Partly because we were unsystematic, partly because homeschooling is illegal and taboo in Sweden and this had worked itself into my body so that I felt revulsion each time I probed the assumption.) In this case, looking at studies and statistics helped remove several needless assumptions. We changed our conclusion (we left Sweden and now homeschool Maud and her sister).

But often the type of problem I like to think about is too personal and messy and qualitative to be resolved cleanly through a statistically significant study. What

do in these situations instead is to consider *counterexamples*.

I like to visualize concrete situations when I make an argument (in the notes for this essay, for example, I continually compare what I say against past writing projects). This makes it easier for me to think clearly. I am tied back into a lived reality, which is rigid, and do not float off into theory, where I have a solid track record of fooling myself. When I have a concrete situation in mind, I can ask myself, “What is a situation where the opposite happened? Why was that?” I can list the characteristics of the situation that inform my conclusion and then systematically look for cases that have other characteristics. In “[Childhoods of exceptional people](#),” for example, I wrote about parenting from the perspective of concrete biographies. The sample was unsystematic. But once I had extracted what I thought were the common patterns, I asked myself, “So whom does this apply to?” Then I added the people that came to mind to the sample and ended with a distribution that was good enough for my purposes.

Counterexamples are useful in two ways. Either you find a counterexample that proves one of the premises wrong but b) does not change your mind about the conclusion. Lakatos calls this a local (and non-global) counterexample. This means there is something wrong with your explanation. Perhaps you need to change that part of the explanation? Or perhaps you can simply drop it, making the mental model simpler and more general? Local counterexamples help you improve your explanation and get a better understanding.

There is a scene in the last season of *Breaking Bad* that illustrates this. The main character, whatever his name was, is a teacher that starts a meth lab. This can be thought of as his conclusion (“I should get into the meth business”) and when asked to defend this decision he explains the claim by saying, “I need to support my family.” This is false. There are better ways for him to do that (he has an old friend who offers him money). That is a local counterexample. In the final season

he admits to himself: “I did it because it made me feel alive.” This doesn’t change his conclusion (he does not change his mind about the meth) but it gives him a deeper and more correct understanding of himself.

Other times, the counterexample you find undermines the whole idea—a *global counterexample*. You explain and stretch your conclusion and discover that one of the premises does not hold up, and there is no way to patch it. The fracture spreads right up to the conclusion. Now—this is what we have been longing for—there is a big hole of confusion where before there was a mental model. It is time to replace it with something more subtle and deep that incorporates the critique.

How to do this, and do it in the most interesting way possible, is the topic of the next part (which I have no idea when I’ll finish).

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Be patient with problems

HENRIK KARLSSON • OCTOBER 18, 2023



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Sara Pendergast Apr 25, 2024 Edited

♥ Liked by Henrik Karlsson

I spent the last 11 weeks drawing the same still life. My marks trying to pin down what I saw. Every day for the first several weeks I'd come to it only to find my marks of the previous day not quite depicting what I saw in front of me. Only after frustratedly re-rendering the piece again and again did I realize my light source changed as the sun rose and fell. I could not hold a constant view of these objects.

So, then, the questions I posed to myself changed. How might I draw something that changes? Everything changes. What about change is important to my drawing? How do changing conditions influence my ability to see? How might I become more attuned to changes as they influence what I am seeing something? What about my reactions to changes as I view something is worth making into a mark? All these thoughts I found to parallel your musings about writing today.

In the end, I am trying to learn a nonverbal language to help me think about life. And, I'm just beginning...

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Celine Nguyen 🌐 Apr 26, 2024

♡ Liked by Henrik Karlsson

I loved reading this, and it's such a beautiful way of making the abstract belief "writing helps you think more clearly!" into a sharper, clearer process. Making one's tentative, early, provisional ideas more concrete in writing ("unfolding" those ideas and beliefs) forces intellectual rigor—and it's a form of rigor that can't really exist as long as things are just sitting in the mind, unactualized and unarticulated.

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